

EcoCollect

Recycling trash, reclaiming value.

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Agenda



Introduction



UI/UX Design (Figma)



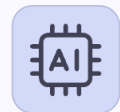
Backend System



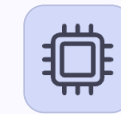
Related Work



Mobile Application (Android)



Machine Learning Model
(Material Detection)



Methodology



Frontend Dashboard (Admin
Web Interface)



Demo



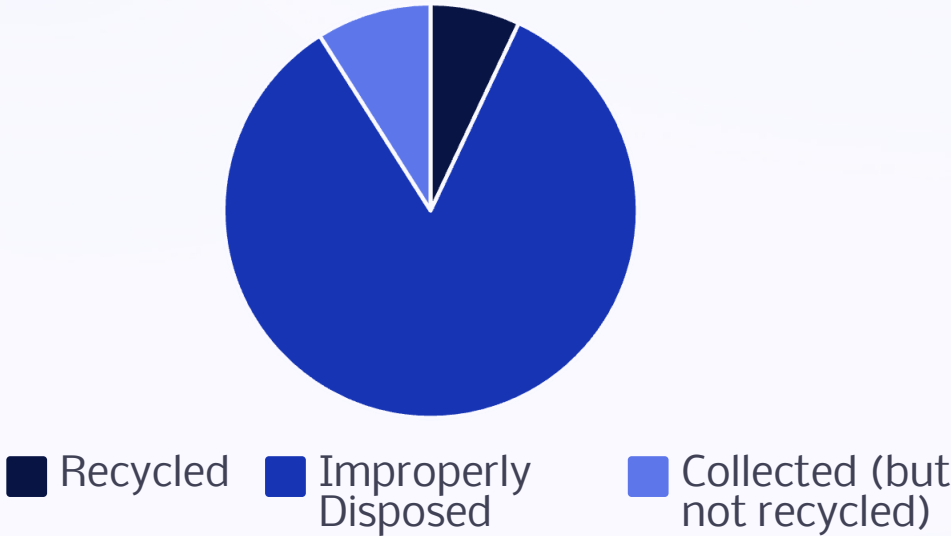
Introduction

Waste is a growing global issue – harming the environment, wasting resources, and affecting communities worldwide. But while many countries are developing new ways to manage waste, recycling efforts in the Middle East and North Africa still face major challenges.

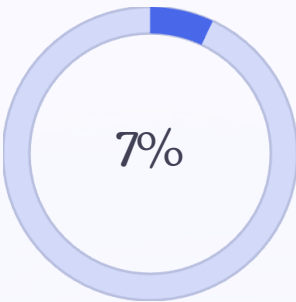
We saw an opportunity to focus on Egypt, where waste continues to pile up, and recycling systems are often inefficient or hard to access. This inspired us to create a tech-based solution that makes recycling easier, more rewarding, and more connected to real impact.

Waste and Recycling Statistics in Egypt

- Egypt produces about 100 million tons of waste every year.
- Each person generates 0.4 to 1 kg of waste daily.
- In cities, waste collection rates range from 40% to 85%.
- In rural areas, rates drop to 0%- 35%.
- Only 7% of waste is actually recycled.
- Around 80%- 88% is disposed of improperly.
- There's a major opportunity: 1.5 million tons of plastic are available for recycling.
- The plastic recycling market is expected to hit \$38 million in 2024.



Tons of waste produced annually



Waste recycled



Tons of plastic available for recycling



Plastic recycling market value in 2024 (est.)

Related Work

- **Bekia (Egypt):** Waste-for-points app (plastic, paper, oil, electronics, etc).
- **D.Waste:** AI identifies waste from photos, users earn points, shows drop-off spots.
- **TrashOut:** Users report illegal dumps, points for tagging, data sent to authorities.



Features

- **Waste Submission by Users:**
Users can submit recyclable materials such as plastic, metal, glass, and cardboard.
- **Collector Matching:**
After the material type is detected, the system suggests nearby registered collectors who buy that material.
- **Transaction Tracking:**
Every transaction (material submission, pickup, and payment) is tracked and stored.
- **Reward Points System:**
Users earn reward points for each successful transaction.
- **Admin Dashboard:**
A secure web interface accessible only by administrators.
- **Analytics & Insights:**
The admin dashboard includes charts and metrics that provide insights into:
 - Most collected waste types.
 - Top-performing collectors.
 - Regions with the highest recycling activity

How It Works

EcoCollect makes recycling easy in just a few steps:

1. Take a Photo Snap or upload a picture of your waste.
2. Get It Identified The app tells you what material it is – plastic, glass, or cardboard.
3. Find a Buyer See nearby buyers or recycling centers who want it.
4. Earn Points You get points for recycling. Trade them in for rewards or cash.
5. Track Everything Admins use a dashboard to manage users and monitor recycling activity.

Methodology

EcoCollect is an end-to-end recycling solution built on five key pillars that work together seamlessly:

1 Mobile App

Our intuitive app guides users through the recycling process, from identifying materials to earning rewards.

2 Web Dashboard

Admins can efficiently manage the platform, monitor analytics, and oversee buyer profiles.

3 Robust Backend

The powerful Laravel-powered API handles authentication, transactions, and background tasks.

4 Machine Learning

Our 85%+ accurate TensorFlow Lite model identifies recyclables in seconds, even offline.

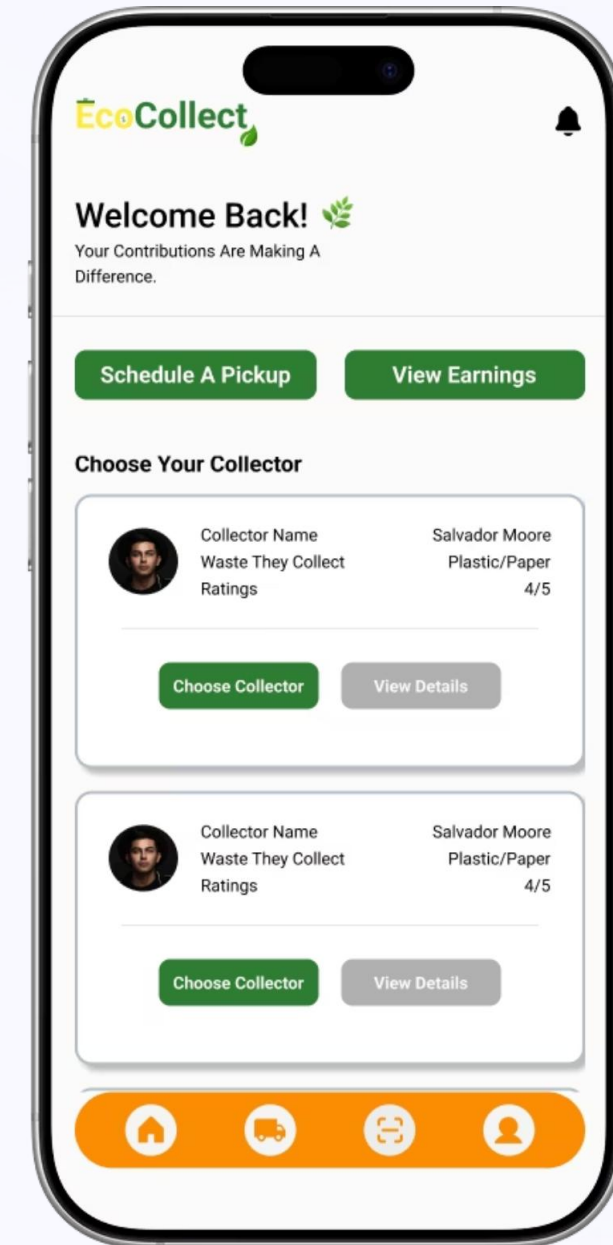
5 Sleek UI/UX

Designed in Figma, the app and dashboard offer a delightful user experience.

UI/UX Design (Figma)

- Designed with **simplicity and clarity** in mind.
- Mobile app: Easy navigation, guides users through recycling → rewards.
- Admin dashboard: Clean visuals, efficient controls, real-time data.
- Built in **Figma** for modern, responsive design.

[Figma Prototype Link](#)





Welcome To EcoCollect!

EcoCollect Connects You With Reliable Services To Collect Your Recyclables, Ensuring You Earn Money While Contributing To A Cleaner Environment. Start Trading Your Waste Today And Make Sustainability Rewarding!

Start Now

Mobile Application (Android)

The EcoCollect mobile application is designed for Android users to easily participate in the recycling process. The app allows users to upload images of recyclable materials, which are then identified using machine learning. It provides a simple and intuitive flow to ensure that even first-time users can quickly navigate the process of selling recyclables.

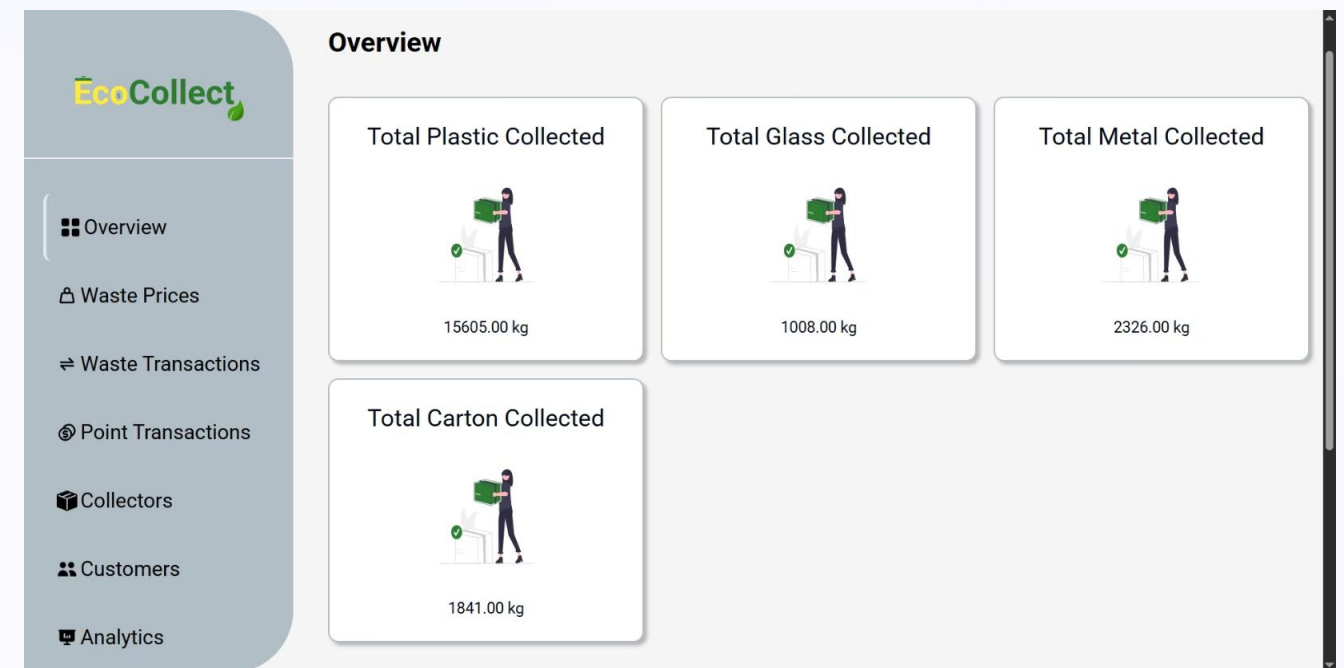
- Framework: Android compose
- Language: Kotlin
- Architecture: MVI

Frontend Dashboard (Admin Web Interface)

The frontend dashboard serves as the control center for administrators to manage users, transactions, and system analytics. It's designed to be highly interactive and responsive, allowing admins to monitor platform activities, view real-time data, and manage buyer profiles.

- Framework: React

Dashboard Live Preview: <https://eco-collect.netlify.app/>



Backend System

The backend system of EcoCollect is built to ensure reliable, secure, and efficient handling of all the app's data and processes. Using the Laravel framework, we've developed a robust and scalable API that powers the entire platform. The backend manages user authentication, handles database transactions, and processes background tasks such as notifications and reward calculations.

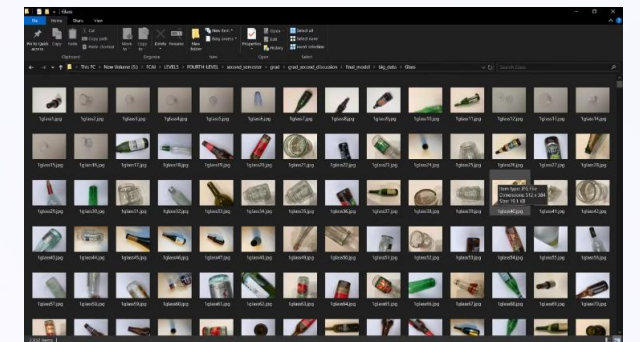
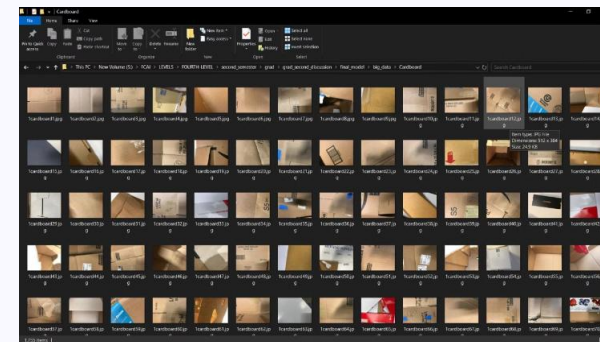
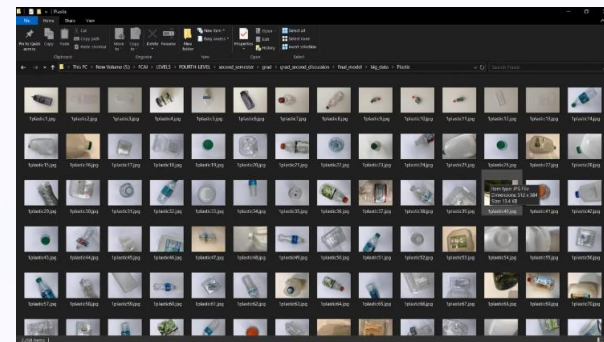
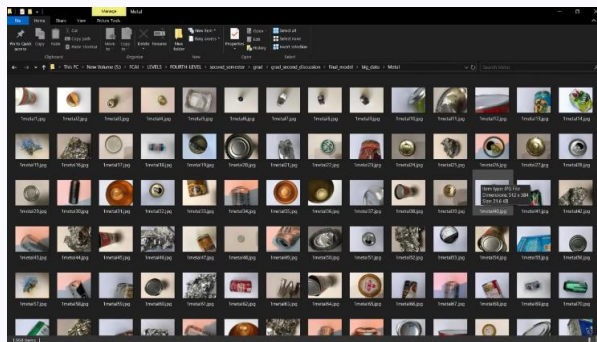
- Framework: Laravel
- Language: PHP
- Database: MySQL

Schema: https://drive.google.com/file/d/1SuDsoIV-G28IBkH2_9aMxGSVfZqqejrP/view?usp=sharing

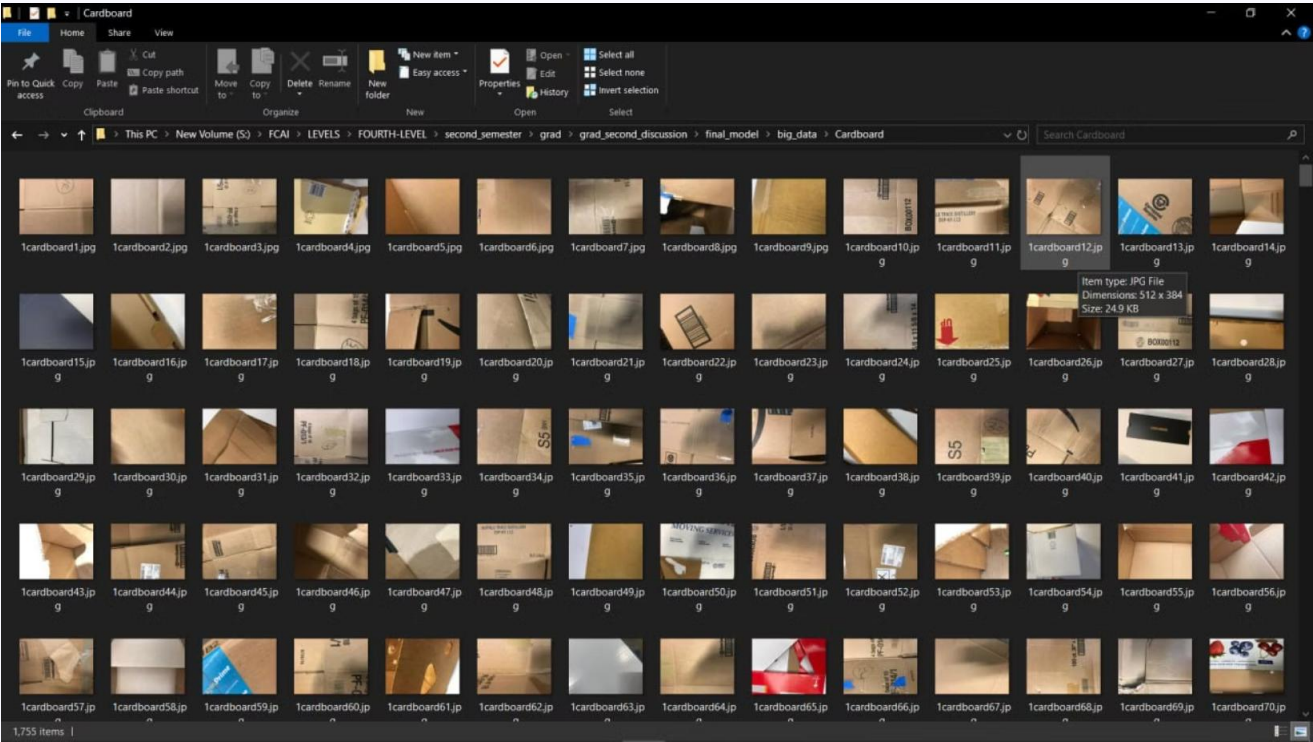
Machine Learning Model (Material Detection)

The machine learning model enables the EcoCollect app to accurately identify recyclable materials from images. Trained with a large dataset of labeled images, the model achieves over {83} % accuracy in material classification. The TensorFlow Lite model runs directly on the mobile device, allowing users to instantly get results without needing an internet connection.

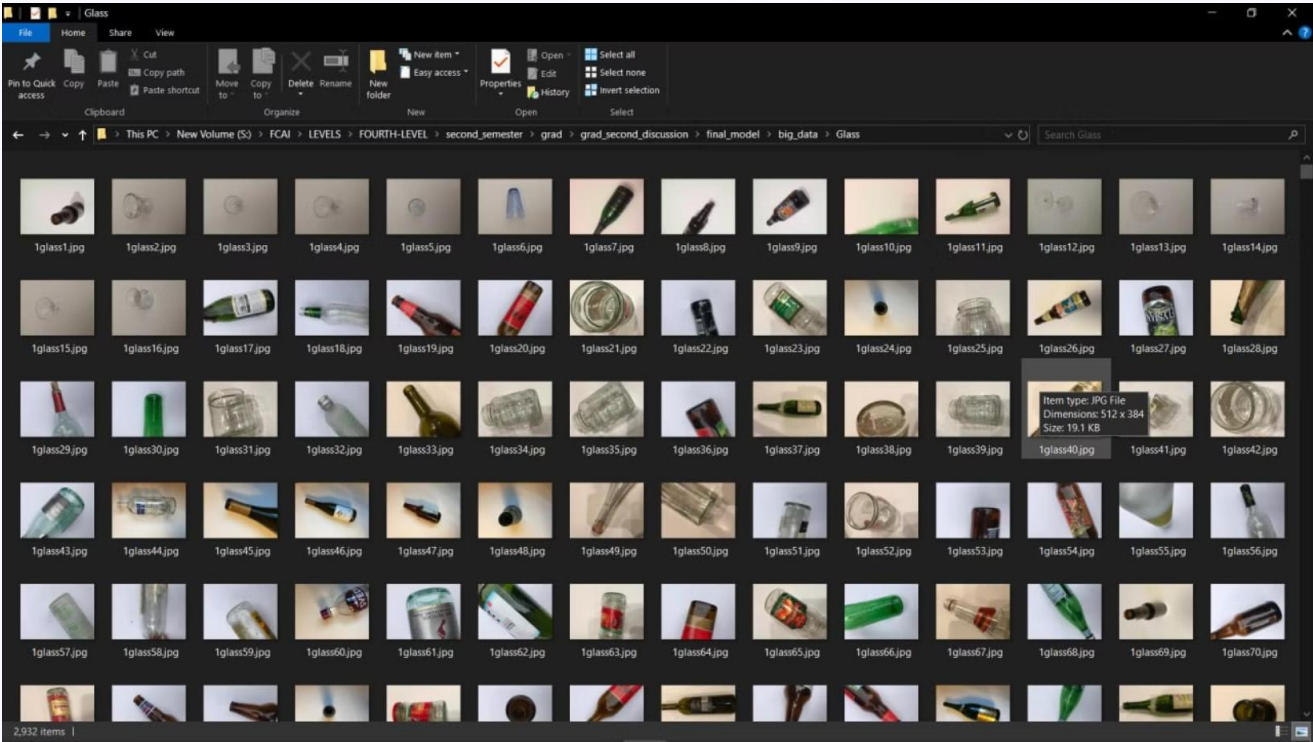
- Framework: TensorFlow / Keras
- Dataset Format: Folder-based classification (/glass, /plastic, /metal, /cardboard)



Dataset Images

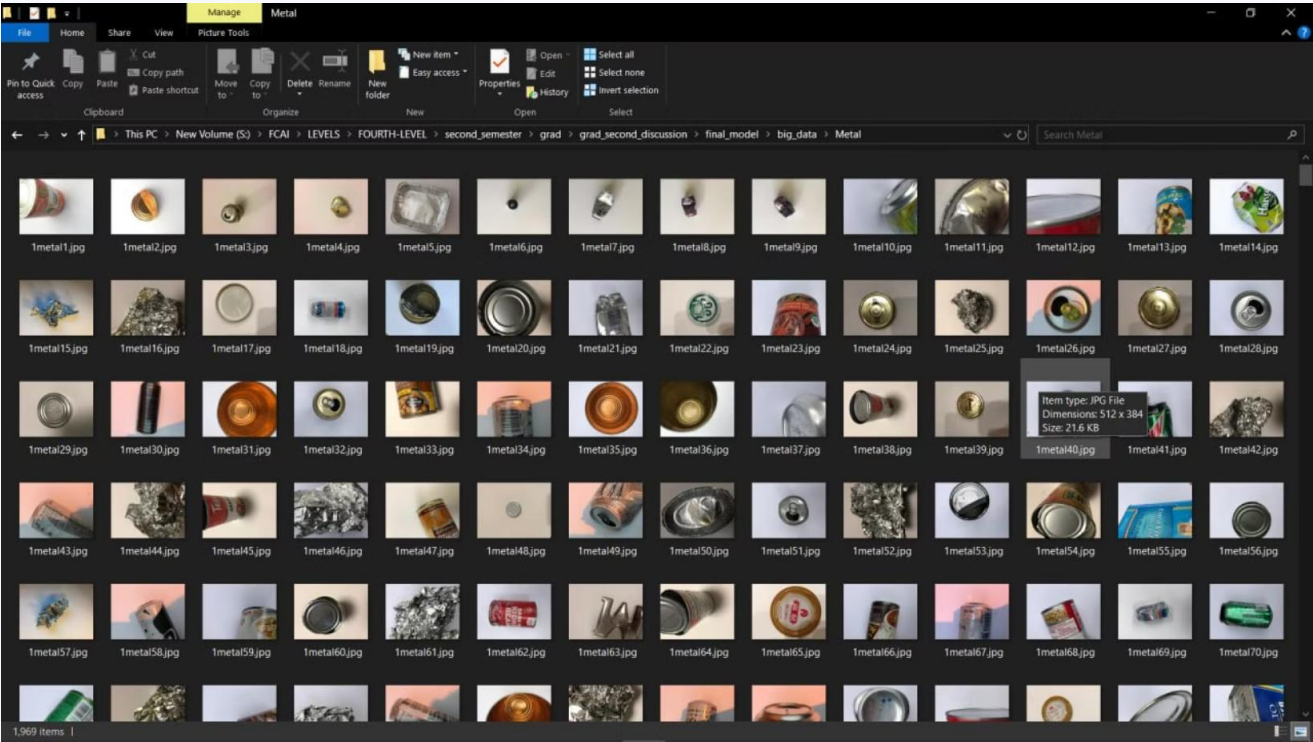


CardBoard

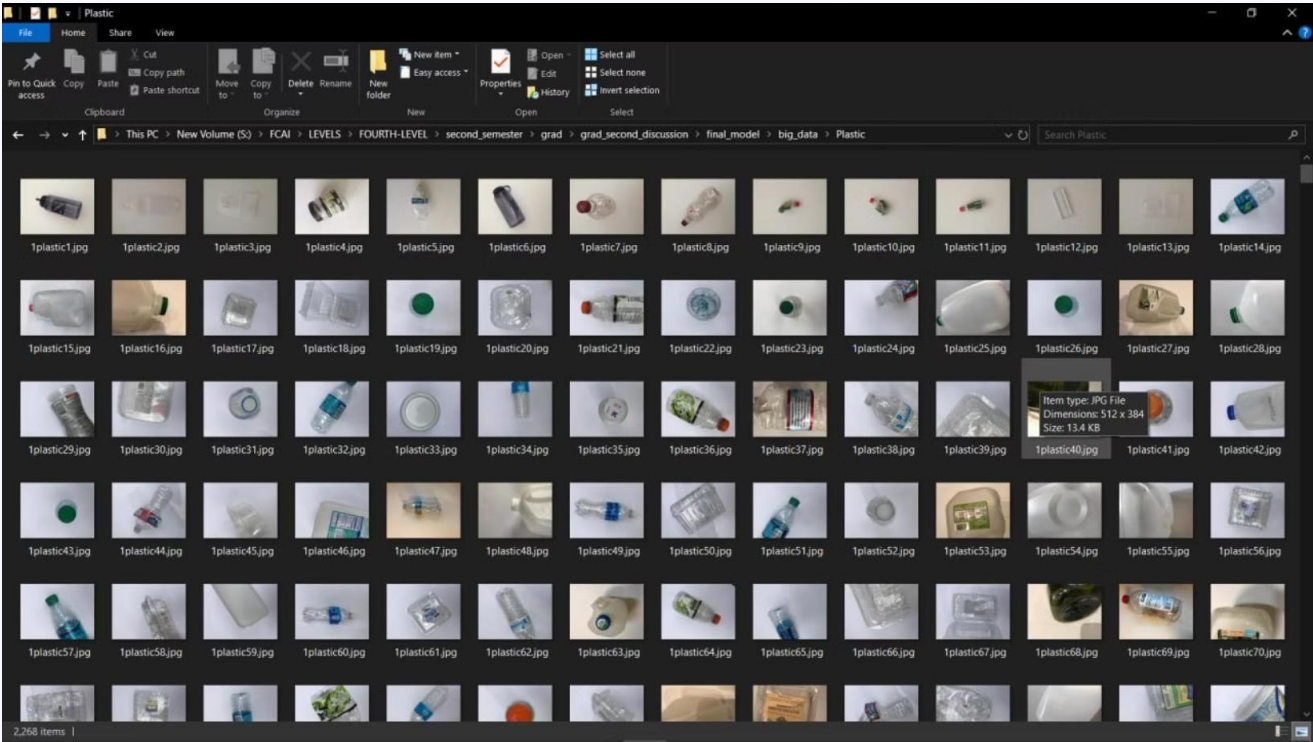


Glass

Dataset Images



Metal



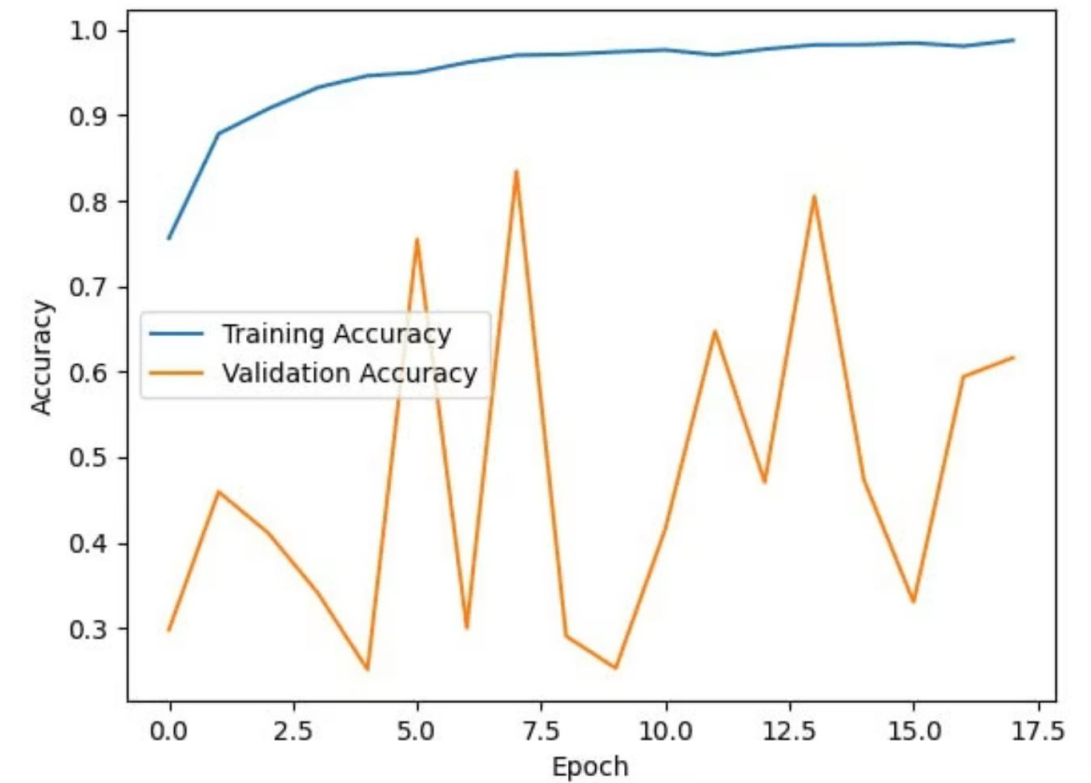
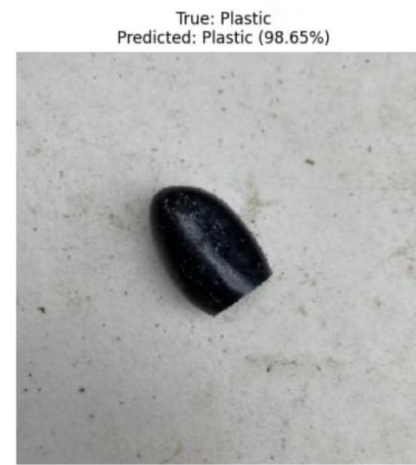
Plastic

Key Training Strategies for EfficientNet-BO

- **Transfer Learning:** We took the pre-trained EfficientNet-Bo and fine-tuned its later layers specifically for our waste categories.
- **Data Augmentation:** To make our model robust and prevent overfitting, we artificially increased our dataset by applying transformations like rotations, brightness changes, and zooms to existing images.
- **Handling Class Imbalance:** Since some waste types might have more images than others, we used **Class Weights** to ensure the model learned equally from all categories.
- **Early Stopping:** This technique prevented our model from overfitting by stopping the training process if its performance on unseen data didn't improve for a certain period.

Model Results

- **Accuracy:** Our model achieved a validation accuracy of approximately **83.62%** in classifying waste images.



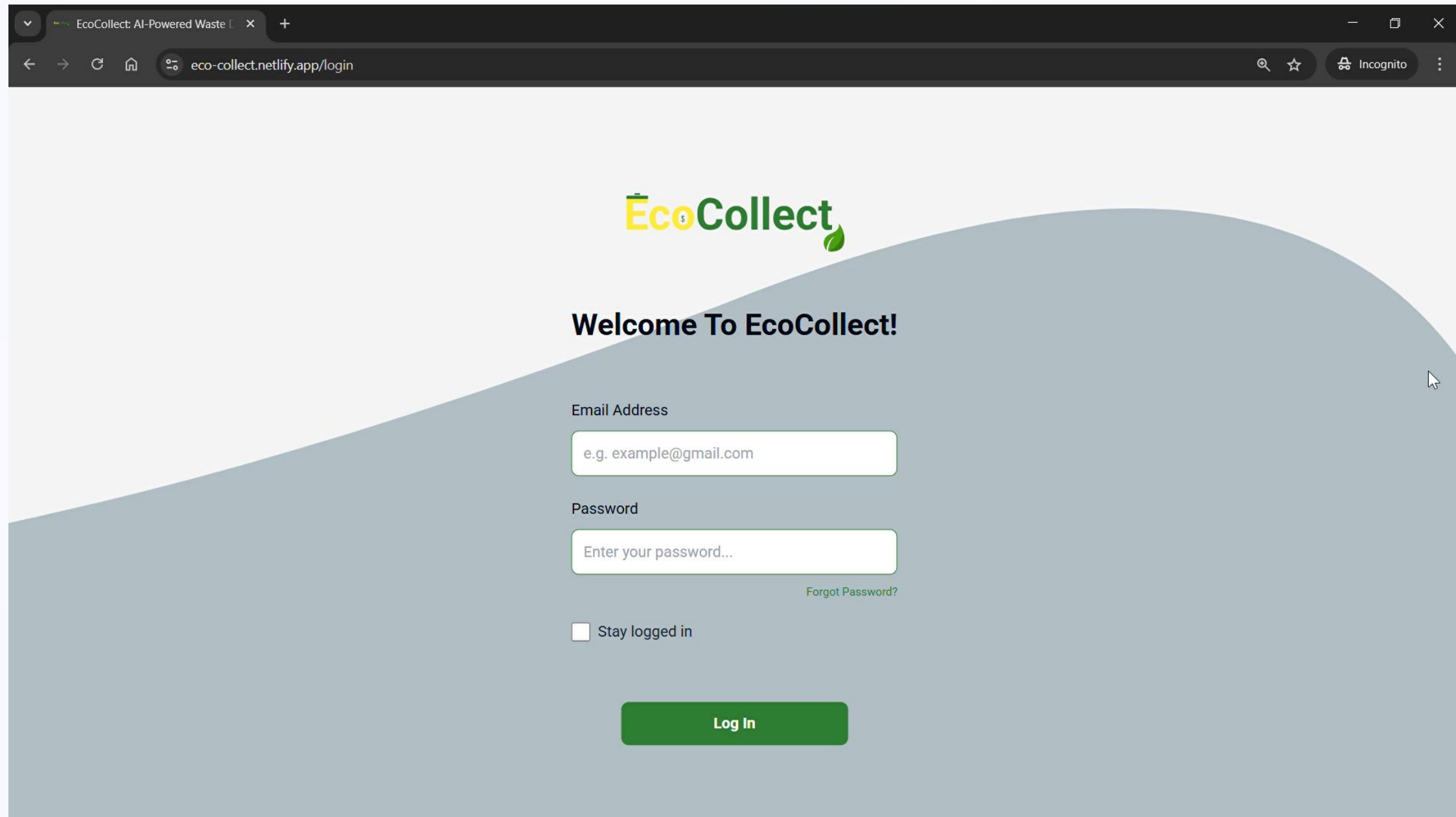
Conclusion

EcoCollect is a smarter way to recycle – fast, simple, and rewarding. With AI material detection, buyer matching, and a powerful admin dashboard, it makes recycling easier for users and scalable for cities. It's not just an app – it's a move toward a cleaner, greener future.



Demo

(Dashbaord)



The screenshot shows a web browser window with the address bar displaying "eco-collect.netlify.app/login". The page features the "EcoCollect" logo at the top, which includes a green leaf icon. Below the logo, the text "Welcome To EcoCollect!" is displayed. The login form consists of two input fields: "Email Address" with a placeholder "e.g. example@gmail.com" and "Password" with a placeholder "Enter your password...". A link "Forgot Password?" is located below the password field. A checkbox labeled "Stay logged in" is positioned below the password field. A green "Log In" button is at the bottom of the form. The background of the page is a light gray with a large, curved, light blue shape on the right side.

EcoCollect

Welcome To EcoCollect!

Email Address

e.g. example@gmail.com

Password

Enter your password...

[Forgot Password?](#)

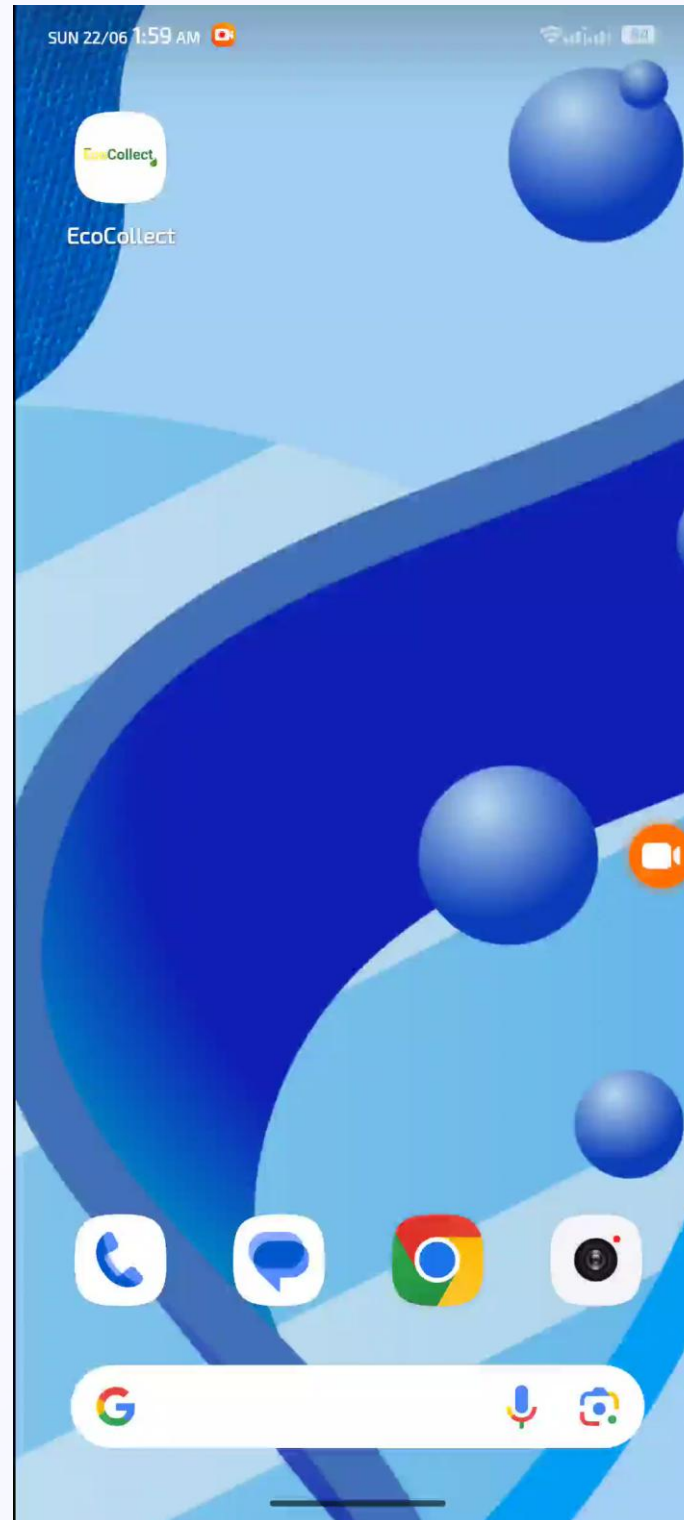
☐ Stay logged in

Log In

Demo

(App)

Customer's Interface



Demo

(App)

Collector's Interface

